

Business Case:

Olist is a marketplace, not a single retailer, it doesn't control the physical goods flow directly (sellers ship independently, often via third-party carriers), yet delivery experience is the thing customers actually feel and rate.

That mismatch (Olist owns the platform and reputation, but not the logistics execution) where no single source system shows the *full* chain from purchase → seller dispatch → carrier handoff → customer doorstep, alongside the *outcome* (review score, repeat purchase).

We will join across `orders`, `order_items`, `sellers`, `customers`, `geolocation` (primary goals) and `reviews` (secondary goal) to see it.

Business Questions for Analysis:

How delivery time affects the business and sales

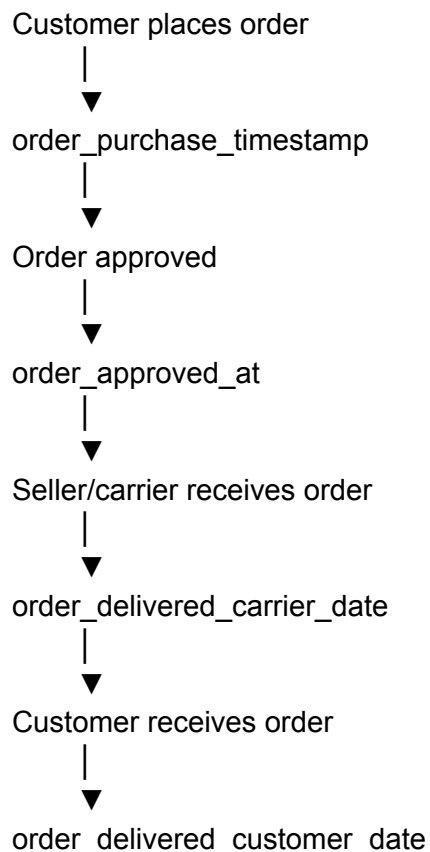
Warehouse location, location of it where can it be to be most efficient

Whether size of product affects the delivery time

Type of product and how it affects delivery time

—

Olist order journey





Expected delivery date
order_estimated_delivery_date

—

1. Are sellers shipping orders late?

This is probably the most obvious one.

We can calculate:

Shipping delay =
order_delivered_carrier_date
-
order_approved_at

For example:

Order	Approved	Carrier	Shipping time
A	1 Jan	2 Jan	1 day
B	1 Jan	5 Jan	4 days
C	1 Jan	10 Jan	9 days

Then we can identify:

Which sellers consistently take longer to hand orders to the carrier?

Business value

The company could:

- identify underperforming sellers
- set seller SLAs
- provide seller performance dashboards
- investigate training/process problems
- potentially prioritize reliable sellers

2. Are customers receiving orders late?

This one is even more directly related to customer satisfaction.

We have:

Actual delivery
vs
Estimated delivery

Calculate:

delivery_delay_days =
actual_delivery_date
-
estimated_delivery_date

Example:

Expected: 10 March

Actual: 14 March

Delay = +4 days

Then classify:

Early

On time

1–3 days late

4–7 days late

8+ days late

This gives you a very executive-friendly KPI:

% of orders delivered late

3. Which sellers have the highest late-delivery rate?

This is where we connect delivery + seller.

For each seller:

Total orders

Late orders

Late delivery %

Average delivery delay

Average shipping time

Example:

Seller	Orders	Late orders	Late %	Avg delay
Seller A	1,200	48	4%	1.2 days
Seller B	800	160	20%	4.8 days
Seller C	500	175	35%	7.1 days

Now you have a seller performance problem that management can act on.

4. Is the seller slow, or is logistics slow?

This is where your analysis can become much more interesting.

Don't just calculate:

"This order was late."

Break the journey into stages.

Stage 1 — Approval

order_purchase_timestamp

↓

order_approved_at

Calculate:

Approval time

approval_delay =

approved_at - purchase_timestamp

Stage 2 — Seller handoff

order_approved_at

↓

order_delivered_carrier_date

Calculate:

Seller/fulfillment delay

seller_handling_time =

carrier_date - approved_at

Stage 3 — Transportation

order_delivered_carrier_date

↓

order_delivered_customer_date

Calculate:

Transit time

transit_time =

customer_delivery - carrier_date

Now we can ask:

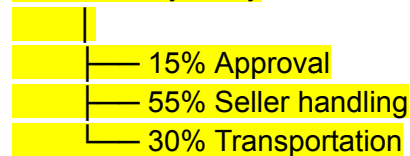
Where is the delay actually happening?

That's much more useful than simply saying "delivery is late."

5. Which stage causes most delays?

Imagine your results are:

Total delivery delay



Your recommendation could be:

"The majority of delivery delays occur before carrier handoff, suggesting seller fulfillment time is a larger contributor than transportation."

That's a business recommendation, not just a data observation.

6. Does late delivery affect customer reviews?

🔥 This could be one of your best analyses.

You have:

Delivery performance
+
Review score

We can compare:

Delivery status	Avg review
Early	4.4
On time	4.2
1–3 days late	3.5
4–7 days late	2.8
8+ days late	2.1

If the actual data shows a strong relationship, you can say:

Late deliveries are associated with lower customer satisfaction.

That's much more compelling to an executive audience.

Important: We should say "associated with," not "causes," unless we do a proper causal analysis.

7. Which sellers are both high-volume AND poor performers?

This is another excellent business case.

You don't necessarily want to focus on a seller with:

3 orders

100% late

That's not necessarily a major business problem.

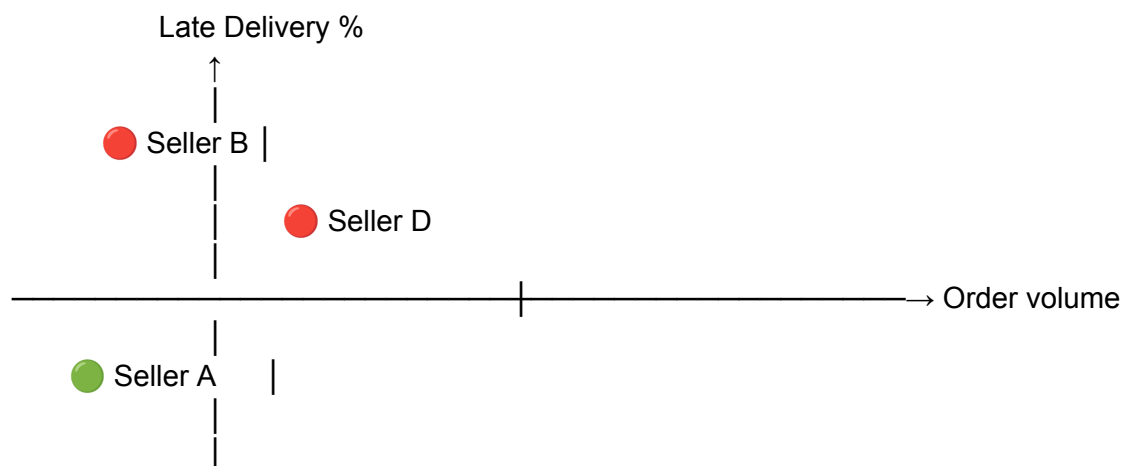
Instead, identify sellers with:

High order volume

+

High late-delivery rate

For example:



This creates a seller risk matrix.

High volume + high delays

🚨 Priority intervention

High volume + low delays

● Good performers

Low volume + high delays

● Monitor

Low volume + low delays

☐ Low priority

That's a great visualization for your final presentation.

*** Considerations for seasonal factors

8. Which product categories experience more delivery problems? (combined with 9)

Now connect:

Product category

+

Delivery performance

For example:

Category	Orders	Late %
Furniture	8,000	18%
Electronics	12,000	11%
Fashion	20,000	7%
Health	10,000	5%

Now management can ask:

Why are furniture orders late more often?

Potential reasons could be size, weight, special handling, seller characteristics, etc.

We shouldn't assume the reason from this data alone, but it gives us a clear area for investigation.

9. Does product size/weight affect delivery?

This is a really nice Olist-specific analysis.

Your product table contains things like:

product_weight_g
product_length_cm
product_height_cm
product_width_cm

We can investigate:

Product characteristics

↓

Delivery time

For example:

Do heavier products take longer to deliver?

Or:

Are large products more likely to arrive late?

You could create weight bands:

< 1 kg

1–3 kg

3–5 kg

5–10 kg

10+ kg

and compare:

Average delivery days

Late delivery %

10. Which states/locations have delivery problems?

You have customer and seller locations.

So we can analyze:

Customer state

↓

Delivery performance

For example:

State	Avg delivery	Late %
SP	8.2 days	7%
RJ	10.4 days	11%
BA	13.1 days	18%

Again, this could help identify geographic/logistics issues.

11. Did the seller approve the order quickly?

There's another subtle business case here.

You asked:

"if seller are approving etc."

The Olist data has order approval, but be careful with the interpretation.

order_approved_at doesn't necessarily mean:

"The seller personally clicked an approve button."

It's better to describe it as:

the time at which the order was approved in the order process.

We can measure:

purchase timestamp



approval timestamp

and calculate:

Approval processing time.

Then investigate:

Are orders with longer approval times more likely to be delivered late?

That's a very reasonable business hypothesis.

★ **What I'd recommend for YOUR project**

Because you only have two weeks, don't try to analyze all of these.

I'd choose 3 connected business questions.

Business Question 1

How well are sellers fulfilling orders?

Metrics:

Average seller handling time

Late shipment %

Seller ranking

High-volume underperforming sellers

Business Question 2

How well is the delivery network performing?

Metrics:

Average delivery time
On-time delivery %
Late delivery %
Delivery delay by state/category
Business Question 3 ★

Does delivery performance relate to customer satisfaction?

Metrics:

Delivery status



Review score

This gives you a really nice story:

SELLER



fulfillment



SHIPPING



delivery performance



CUSTOMER



satisfaction



REVIEW SCORE

That's much more compelling than a project that only says:

"Category X sold the most products."

And this affects our warehouse design

This is why I wanted you to understand facts vs dimensions before we build the warehouse.

Our fact_orders should contain the dates needed to calculate these metrics:

fact_orders

order_id

customer_key

order_purchase_at

order_approved_at

order_delivered_carrier_at

order_delivered_customer_at
order_estimated_delivery_at

approval_delay_days
seller_handling_days
transit_days
total_delivery_days
delivery_delay_days
delivery_status

Then we can connect it to:

dim_customer
dim_date

And connect fact_order_items to:

dim_product
dim_seller

And eventually connect reviews for the satisfaction analysis.

My recommendation

Yes, let's make delivery performance one of the main business themes of your project.

It gives you a much stronger executive story:

"We built an e-commerce analytics warehouse to identify sales trends and uncover operational bottlenecks in order fulfillment and delivery."

Then your final presentation can show:

Revenue → Seller Performance → Delivery → Customer Satisfaction

That's a coherent business narrative rather than a collection of unrelated charts.